

Clinical Summary Generation Using an Open-Source LLM

Problem statement: Build a clinical-summary generation workflow for input call transcripts.

Constraints

1. Use English-language synthetic call transcripts only.
2. Use the specified open-source [Gemma 12B model](#). Document the model version, inference configuration, hardware, and rationale for model selection. We can provide a GPU if needed.
3. Timeline: 5 days.
4. The system must run locally or on a provided GPU environment.

Scenario: Build and evaluate an open-source LLM workflow that converts a transcript into a structured clinical summary. The workflow must produce summaries that are clinically useful, source-grounded, and safe for nurse review. Example transcripts and reference summaries are provided separately.

The clinical summary must contain:

1. Chief Complaint: Caller, patient, relationship, reason for call, and relevant context.
2. Assessment: Symptoms, duration, severity, medication or supply concerns, and relevant caller-reported observations.
3. Response: Actions stated by the nurse, such as medication refill requests, nurse visits, escalation, or team follow-up.
4. Education: Instructions, return precautions, and follow-up guidance communicated during the call.

Safety and Quality Rules

1. Do not invent patient identity, symptoms, medications, doses, actions, or outcomes.
2. Attribute statements correctly, for example, "Caller reported..." or "Nurse stated..."
3. Preserve uncertainty. If a medication name, dose, identity detail, or symptom is unclear because of ASR errors or caller uncertainty, mark it as unclear rather than guessing.
4. Do not convert a planned action into a completed action. Use wording such as "planned to notify the nurse" when appropriate.
5. Detect transcripts that do not contain enough clinical content and return Not Applicable.
6. Flag high-risk content for nurse review, including uncontrolled symptoms, medication concerns, suicidal statements, breathing concerns, or escalation requests.

Tasks

1. Build a clinical-summary generation pipeline using the open-source LLM and deterministic validation logic.
2. Create a simple interface or API that accepts a transcript and returns a structured clinical summary.
3. Build a synthetic evaluation dataset containing routine calls, ambiguous calls, ASR-error cases, medication mentions, supply requests, and high-risk escalation cases.
4. Develop an automatic evaluation method and report the following metrics:
 - Summary completeness and factual accuracy
 - Identity accuracy
 - Medication-name and dose accuracy
 - Time to first token and total response time
5. Achieve at least 95% accuracy on the defined automatic evaluation metric. Clearly state the metric definition, evaluation dataset size, and failure cases.
6. Create synthetic SFT or LoRA training data to improve clinical-summary quality.

7. Reduce end-to-end clinical-summary latency to p95 below 15 seconds on the stated evaluation hardware.
8. Compare and report:
 - Baseline performance without fine-tuning
 - Fine-tuned performance using SFT or LoRA
 - Baseline and fine-tuned latency
 - Optimization methods used to reduce latency
9. Document methods used to improve safety, accuracy, consistency, and latency.

Submission

Submit one ZIP folder containing:

1. `code/` runnable Python code
2. `data/` synthetic training, validation, and evaluation datasets
3. `outputs/` generated summaries, evaluation reports, and latency results
4. `README.md` installation and run instructions
5. `report.pdf` or `report.md` pipeline design, model selection, safety controls, evaluation methodology, baseline versus fine-tuned results, latency analysis, and limitations
6. A short demo video showing transcript input, clinical-summary generation, and safety or review flags.